Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2023/2024

COURSE NAME	:	DATABASE
COURSE CODE	:	BIC 21404
PROGRAMME CODE	:	BIS
EXAMINATION DATE	:	JANUARY/FEBRUARY 2024
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

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Q1 (a) What is data normalization, and why is it important in database analysis?

(5 marks)

(b) Explain the relevance of data warehouse in today's context.

(5 marks)

(c) How enterprise data are related to database?

(4 marks)

Q2 **Table Q2.1** shows lists of dentist and patient appointment data. A patient is given an appointment at a specific time and date with a dentist. On each day of patient's appointments, a dentist is allocated to a specific surgery ID for that day.

Table Q2.1

staffNo	**dentistName**	**patNo**	**patName**	**appointment**		**surgeryID**
				date	time	
S1011	Tony Smith	P100	Gillian White	12-Sep-01	10:00	S15
S1011	Tony Smith	P105	Jill Bell	12-Sep-01	12:00	S15
S1024	Helen Pearson	P108	Lan MacKay	12-Sep-01	10:00	S10
S1024	Helen Pearson	P108	Lan MacKay	14-Sep-01	14:00	S10
S1032	Robin Plevin	P105	Jill Bell	14-Sep-01	16:30	S15
S1032	Robin Plevin	P110	Johon Walker	15-Sep-01	18:00	S15

(a) Based on the **Table Q2.1**, provide **ONE (1)** example of insertion, deletion, and modification anomalies.

(6 marks)

(b) Write the relational schema and draw its dependency diagram. Identify all dependencies, including a transitive dependency.

(15 marks)

(c) Perform data normalization process to meet the Third Normal Form (3NF) requirements

(15 marks)

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Q3 Answer **Q3(a) – Q3(c)** based on **Figure Q3.1**

Table name: EMPLOYEE

Emp#	Emp_title	Emp_name	Emp_Initial	Emp_DOB	Store_Code
1	Mr.	Haris	M	21/5/1980	3
2	Ms.	Hana	N	13/4/1985	2
3	Ms.	Farah	J	12/9/1989	4
4	Mrs.	Rozy	M	21/10/1989	5
5	Mr.	Smith	F	1/8/1988	3
6	Mr.	Hannan	K	12/2/1990	1
7	Mr.	Fendi	A	10/7/1987	3
8	Ms.	Kelly	L	17/11/1985	1
9	Mr.	Harun	A	23/3/1986	2
10	Mrs.	Rebecca	P	10/1/1984	4
11	Mr.	Salim	H	12/12/1990	3

Table name: STORE

Store_Code	Store_name	Store_Sales	Region_Code	Emp#
1	Linc	1003455.76	2	8
2	Junction	1421987.39	2	12
3	Fourfront	986783.22	1	7
4	Tupleview	944568.56	2	3
5	Keypoint	29300098.45	1	15

Table name: REGION

Region_Code	Region_Desc
1	East
2	West

Figure Q3.1

(a) Identify the primary key and the foreign key for each table.

(4 marks)

(b) Do all tables exhibit entity integrity? Justify your answer.

(4 marks)

(c) Do all tables exhibit referential integrity? Justify your answer.

(4 marks)

Q4 The following are the attributes of a STAFF entity:

S_ID, S_ADDRESS, S_QUALIFICATION, S_DATE_OF_BIRTH, S_AGE

Identify:

(a) Composite attribute.

(b) Derived attribute.

(c) Multivalued attribute.

(3 marks)

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Q5 In database processes design:

(a) Distinguish between top-down and bottom-up approaches.

(4 marks)

(b) Distinguish between centralized and decentralized conceptual database design.

(4 marks)

Q6

UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company of information system. The attributes of Shipped items are item ID (unique), weight, dimensions, insurance amount, destination, and final delivery date. The Shipped will be send to a single retail centre. the attributes of retail centre are type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation methods (i.e., flights, truck deliveries). the attributes of the transport are a unique TransportID, a type (e.g, flight, truck).

(a) Create an Entity Relationship diagram that captures this information about the UPS system including the relationship attributes. Be certain to indicate identifiers and cardinality constraints.

(15 marks)

(b) Based on **Figure Q6.1**, write a Structured Query Language (SQL) statement for **Q6(b)(i) – Q6(b)(iv)**.

```
emp(eno, ename, bdate, title, salary, dno)
proj(pno, pname, budget, dno)
dept (dno, dname, mgreno)
workson (eno, pno, resp, hours)
```

Figure Q6.1

(i) The employees (name only) in department 'D1' ordered by descending salary.

(3 marks)

(ii) The project name, department name, and budget for all projects with a budget < RM50,000.

(3 marks)

(iii) The employee numbers and salaries of all employees in the 'Consulting' department ordered by descending salary.

(3 marks)

(iv) The employee's name, project name, employee title, and hours for all works on records.

(3 marks)

- END OF QUESTIONS -

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